

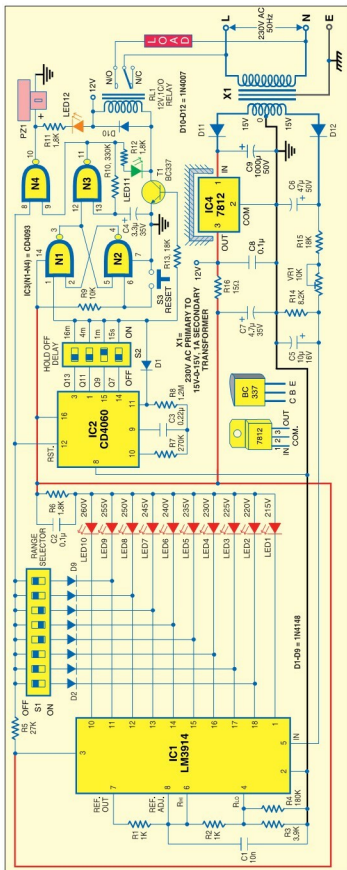


Fig. 1: Circuit of mains monitor

lay of around one minute to disable the clock circuit of CD4060 via diode D1. As a result, the counter stops counting with the selected output held high and remains in this condition until it is reset by the OR gate output. The high output of DIP switch S2 energises relay RL1 via resistor R13 and transistor T1, and the AC mains source connects to the load through relay contacts.

The high output of DIP switch S2 is also applied to the set input of the event memory latch—a negative-edge-triggered flip-flop made of NAND gates N1 and N2. This flip-flop can be reset by pressing memory reset switch S3.

The output of the flip-flop at pin 3 of gate N1 is used to control the oscillator built around NAND gate N3, resistor R10 and capacitor C4. The

frequency of oscillations is around 1 Hz. The oscillator is normally in disabled state as the collector of transistor T1 is low when the mains voltage is within the acceptable range (indicated by glowing of LED11).

When AC mains is within the acceptable range (set by using DIP switch S1), the OR gate output is low, gate N4 output is high and LED12 is off. In short, when the OR gate output is low, the event memory flip-flop resets and relay RL1 energises to connect mains power to the load.

When the mains voltage goes outside the acceptable range, the resulting high level on the OR gate resets IC2 (making all of its outputs low). Transistor T1 cuts off and relay RL1 de-energises to disconnect the load from the mains and set the event flip-flop. The high output of the flip-flop enables the oscillator, which provides 1Hz clock pulse to gate N4. LED12 flashes when the gate is enabled by the OR gate output.

LED11 is off, as the collector of transistor T1 is high. LED12 flashes and LED11 remains off until the mains voltage returns to the acceptable value. When mains power returns to the acceptable voltage, LED12 is disabled and timer IC2 enabled. IC2 starts oscillating to advance the counter. The count keeps incrementing until timer IC2 output goes high (selected by DIP switch S2) after the hold-off delay (time taken to connect the load to mains power after mains power returns to the acceptable voltage).

Output Q9 of CD4060 goes high after hold-off delay of one minute and most of the circuit returns to normal state as detailed above. However, setting the event flip-flop during mains voltage outside the acceptable range leaves the oscillator enabled. As a result, LED11 flashes at 1 Hz, which also indicates that a mains fluctuation has occurred. Switch S3 resets the flip-flop, returning the circuit to its normal state. Piezobuzzer PZ1 connected to the output of gate N4 sounds when mains supply goes out of the acceptable vol-